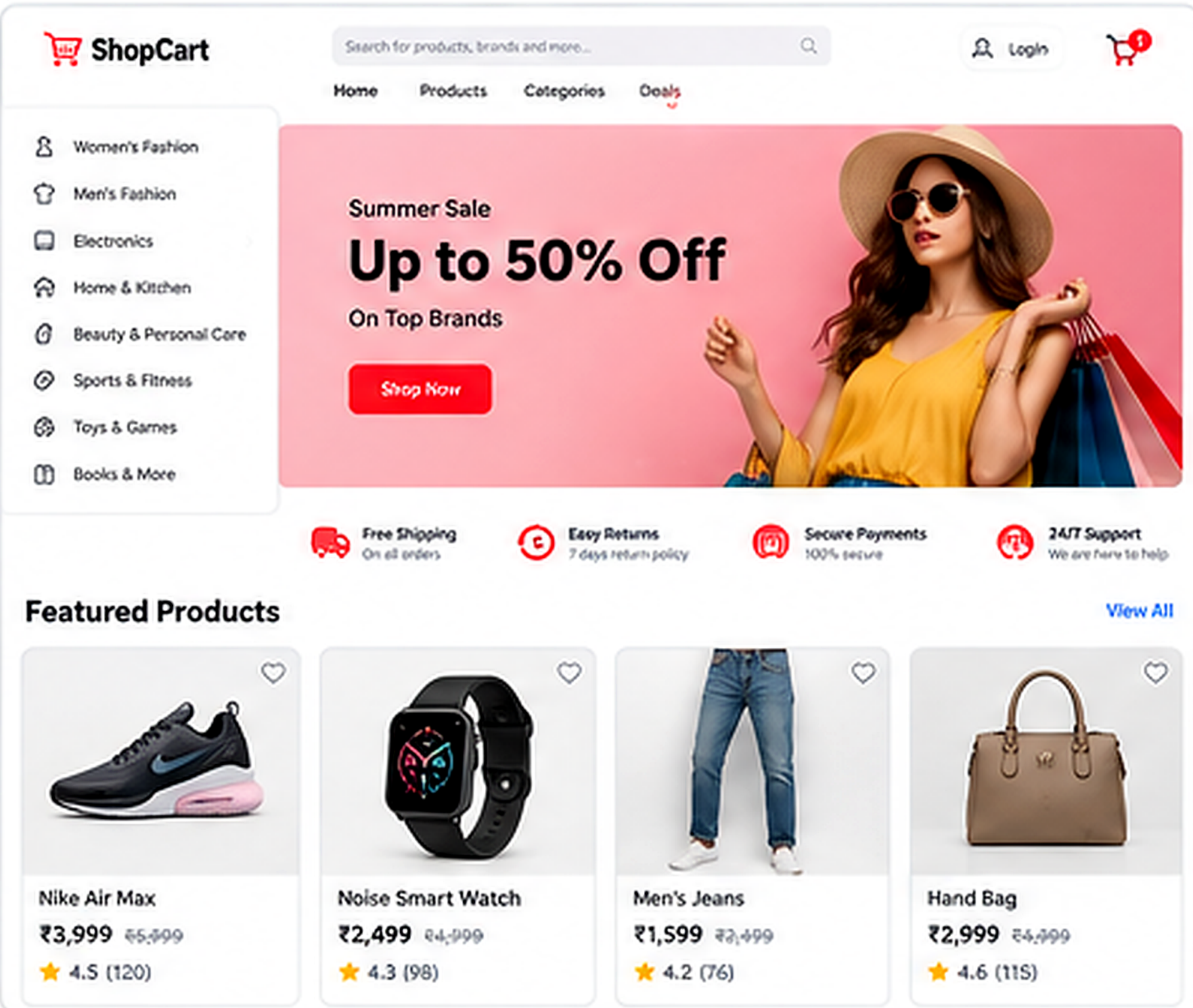


E-Commerce Application

Project Overview & Learning Goals

In this session, we will plan and start building a complete E-Commerce frontend application using React. We will break the application into small, reusable components and understand how these components come together to form full pages.



What will we learn today?

- ✓ Understand the complete application we are going to build
- ✓ Identify pages and their purpose
- ✓ Break each page into reusable components
- ✓ Create project folder structure
- ✓ Build our first reusable component (ProductCard)

By the end of this class you will be able to:

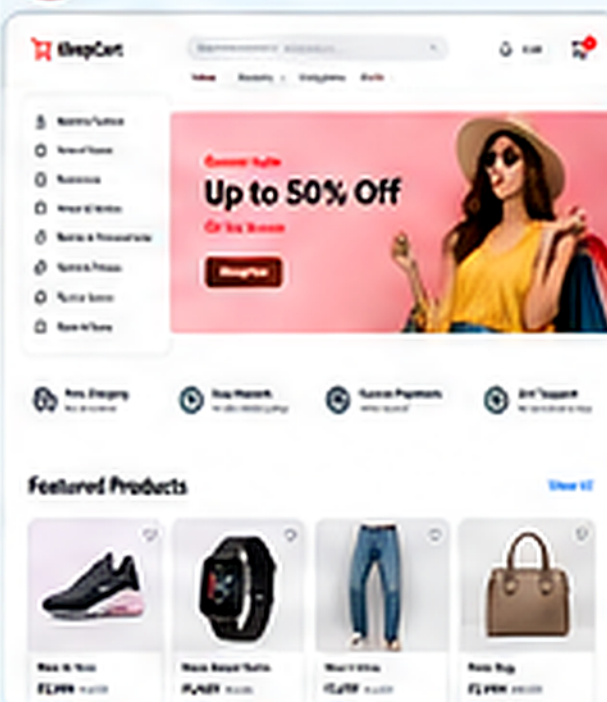
- ✓ Visualize a complete e-commerce application
- ✓ Identify and list the required components
- ✓ Understand the relationship between components and pages
- ✓ Create a proper project structure
- ✓ Build a simple ProductCard component

Application Pages

Main Pages of the E-Commerce Application

Our application will have the following main pages. Each page uses different components to provide specific functionality for the user.

1 Home Page

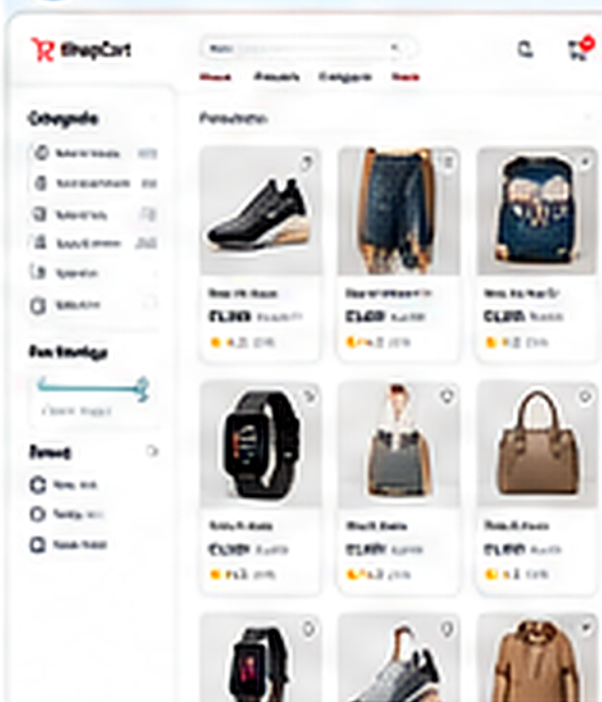


Purpose: Show banners, featured products and categories.

Main Components:

- Navbar
- CategoryList
- ProductList (ProductCard)
- Footer

2 Products Page

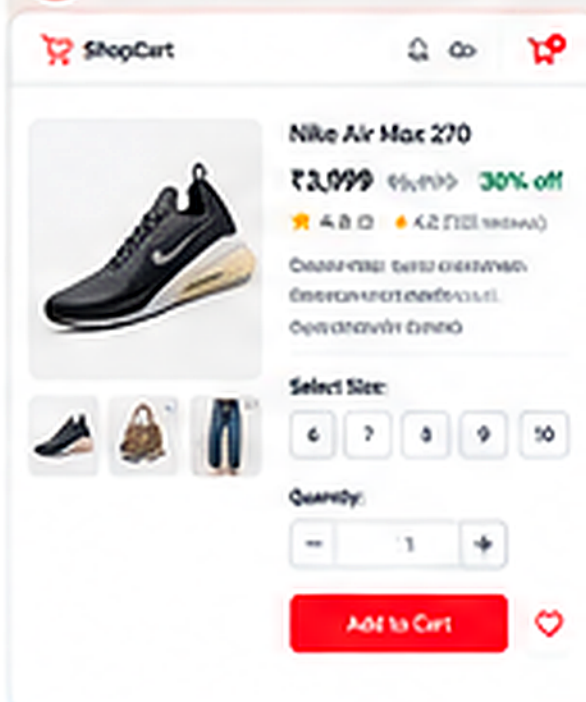


Purpose: Show all products with filters and search.

Main Components:

- Navbar
- Sidebar (Filters)
- ProductList (ProductCard)
- Pagination
- Footer

3 Product Details Page

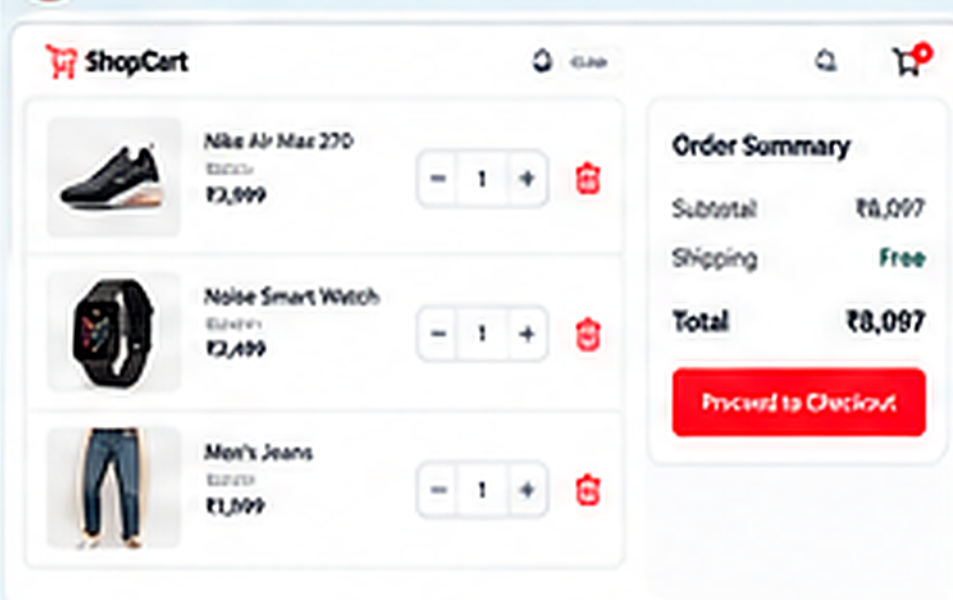


Purpose: Show complete information about a single product.

Main Components:

- Navbar
- ProductDetails
- RelatedProducts (ProductCard)
- Footer

4 Cart Page

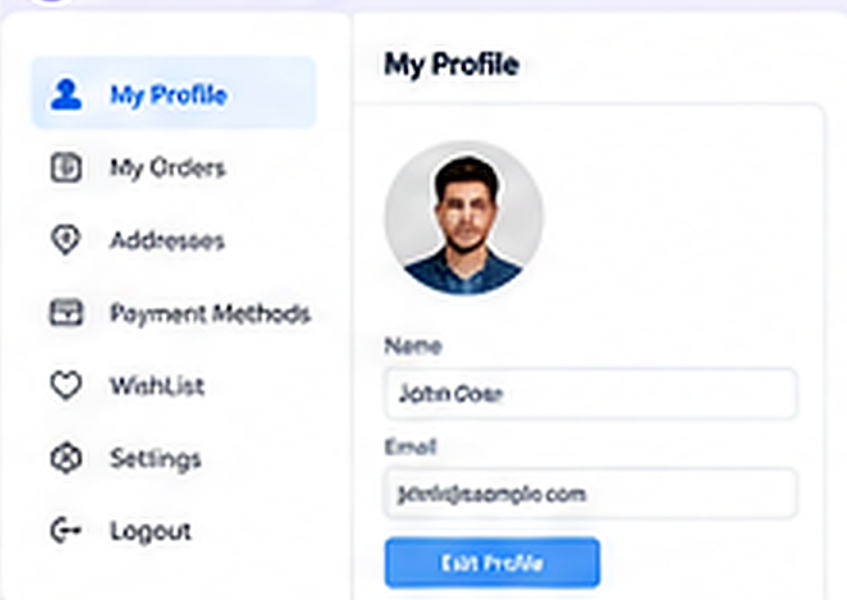


Purpose: Show selected items, update quantity and proceed to checkout.

Main Components:

- Navbar
- CartItem
- CartSummary

5 Profile Page



Purpose: Show and manage user information, orders and settings.

Main Components:

- Navbar
- ProfileCard
- Sidebar (Profile Menu)

Individual Components

Common Components Used in the Application

These are the reusable components that we will build and use across different pages.

1 Navbar



Purpose: Shown on all pages. Contains logo, navigation links, search bar and cart icon.

2 SearchBar



Purpose: Allows users to search for products.

3 Sidebar / Filters

Categories

- ☐ Men's Fashion (120)
- ☒ Women's Fashion (98)
- ☐ Electronics (76)
- ☐ Home & Kitchen (54)

Price Range



Brand

- ☐ Nike (34)
- ☐ Adidas (28)

Purpose: Filter products by category, price, brand etc.

4 Product Card



Nike Air Max 270

₹3,999 ~~₹6,999~~

★ 4.5 (120)

Add to Cart

Purpose: Displays product image, name, price, rating and action button.

5 Cart Item



Nike Air Max 270

₹3,999

Size: 8 Color: White



Purpose: Shows individual item in the cart with quantity controls.

6 Cart Summary

Order Summary

Subtotal ₹8,097

Shipping Free

Total ₹8,997

Proceed to Checkout

Purpose: Shows total price and checkout action.

7 Profile Card



John Doe

john@example.com

Edit Profile

Purpose: Displays user information on profile page.

8 Footer

ShopCart

Customer Care

Help Center

Returns

Track Order

Company

About Us

Careers

Contact



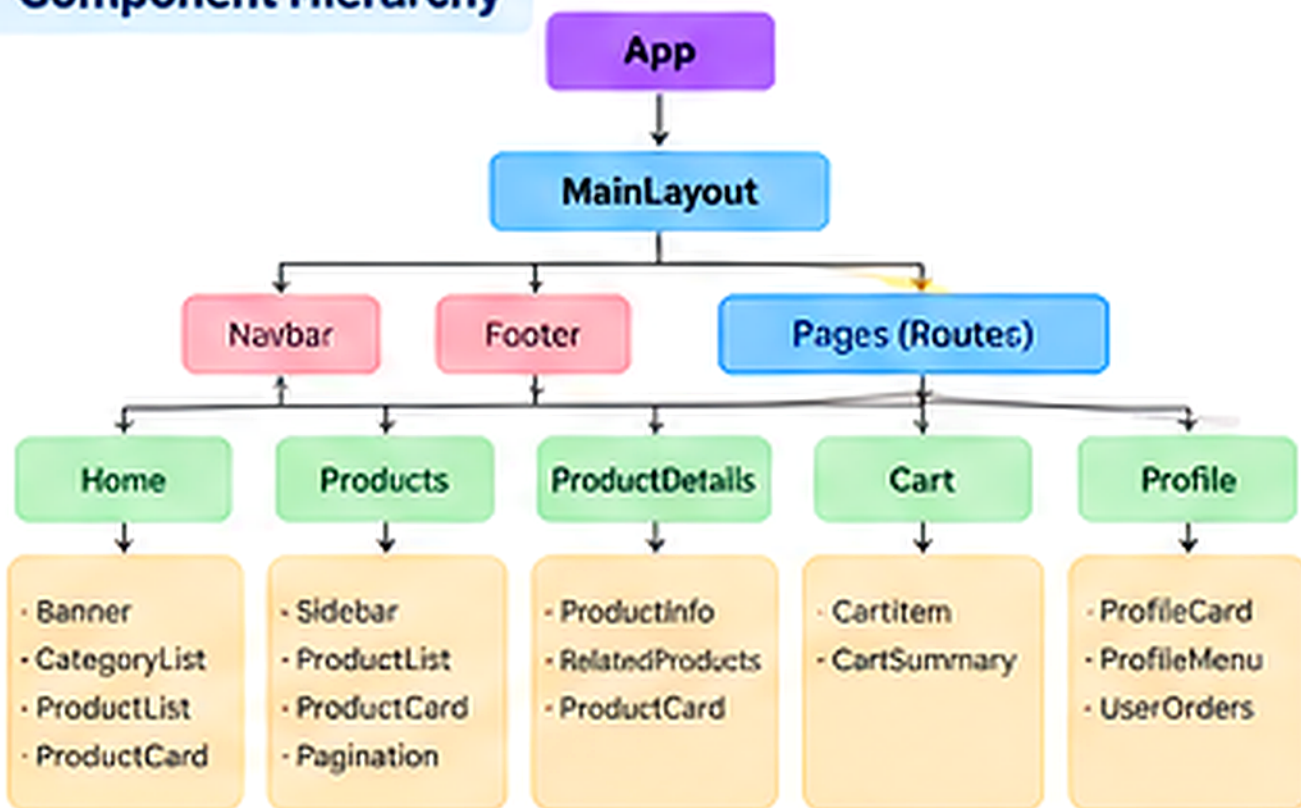
© 2024 ShopCart. All rights reserved.

Purpose: Shown on all pages. Contains important links and company information.

Component Architecture & Project Structure

How Components Combine to Build the Application

Component Hierarchy



Folder Structure (Recommended)

```
src/
├── assets/
├── components/
│   ├── Banner.jsx
│   ├── Footer.jsx
│   ├── ProductCard.jsx
│   ├── ProductList.jsx
│   ├── Sidebar.jsx
│   ├── CartItem.jsx
│   ├── CartSummary.jsx
│   ├── ProfileCard.jsx
│   └── SearchBar.jsx
├── pages/
│   ├── Home.jsx
│   ├── Products.jsx
│   ├── ProductDetails.jsx
│   ├── Cart.jsx
│   └── Profile.jsx
├── layouts/
│   └── MainLayout.jsx
└── App.jsx
    └── main.jsx
```

Key Points

- 1 Components are small, reusable UI parts.
- 2 Pages are complete screens built using components.
- 3 Layout is used for common UI (Navbar, Footer).
- 4 Each component has a single responsibility.
- 5 We pass data using props to make components reusable.
- 6 Same components can be used in multiple pages.
- 7 Folder structure helps to keep the code organized.

Practical Exercise: Build a ProductCard Component

Let's build our first reusable component.

Design Reference



Nike Air Max 270

₹3,999 ~~₹5,999~~

★ 4.5 (120)

🛒 Add to Cart

Step-by-Step Implementation

ProductCard.jsx

```
import React from 'react';

const ProductCard = ({ product }) => {
  return (
    <div className="product-card">
      <img src={product.image} alt={product.title} />
      <h3 className="title">{product.title}</h3>
      <p className="price">₹{product.price}</p>
      <div className="rating">★ {product.rating}</div>
      <button className="add-btn">Add to Cart</button>
    </div>
  );
};

export default ProductCard;
```

Sample Product Data

```
const sampleProduct = {
  id: 1,
  title: "Nike Air Max 270",
  price: 3999,
  image:
    "https://images.unsplash.com/photo-1600183165483-26d7d8eae9",
  rating: 4.5,
};
```

Requirements

- ✓ Display product image, title, price and rating.
- ✓ Show an "Add to Cart" button.
- ✓ Accept product data using props.
- ✓ Use CSS for styling (you can use your existing CSS knowledge).
- ✓ Make the component reusable for different products.

Student Checklist

- ☐ Create ProductCard.jsx file.
- ☐ Use props to receive product data.
- ☐ Display all required information.
- ☐ Add basic CSS styles.
- ☐ Test with multiple sample products.

Next Steps (Homework)

- 1 Plan the remaining components (Navbar, Footer, etc.).
- 2 Think about the layout for each page.
- 3 Try to build one more small component (e.g., Navbar).
- 4 Be ready to integrate components and pages in the next class.